Your First Visualization

PSY 410: Data Science for Psychology

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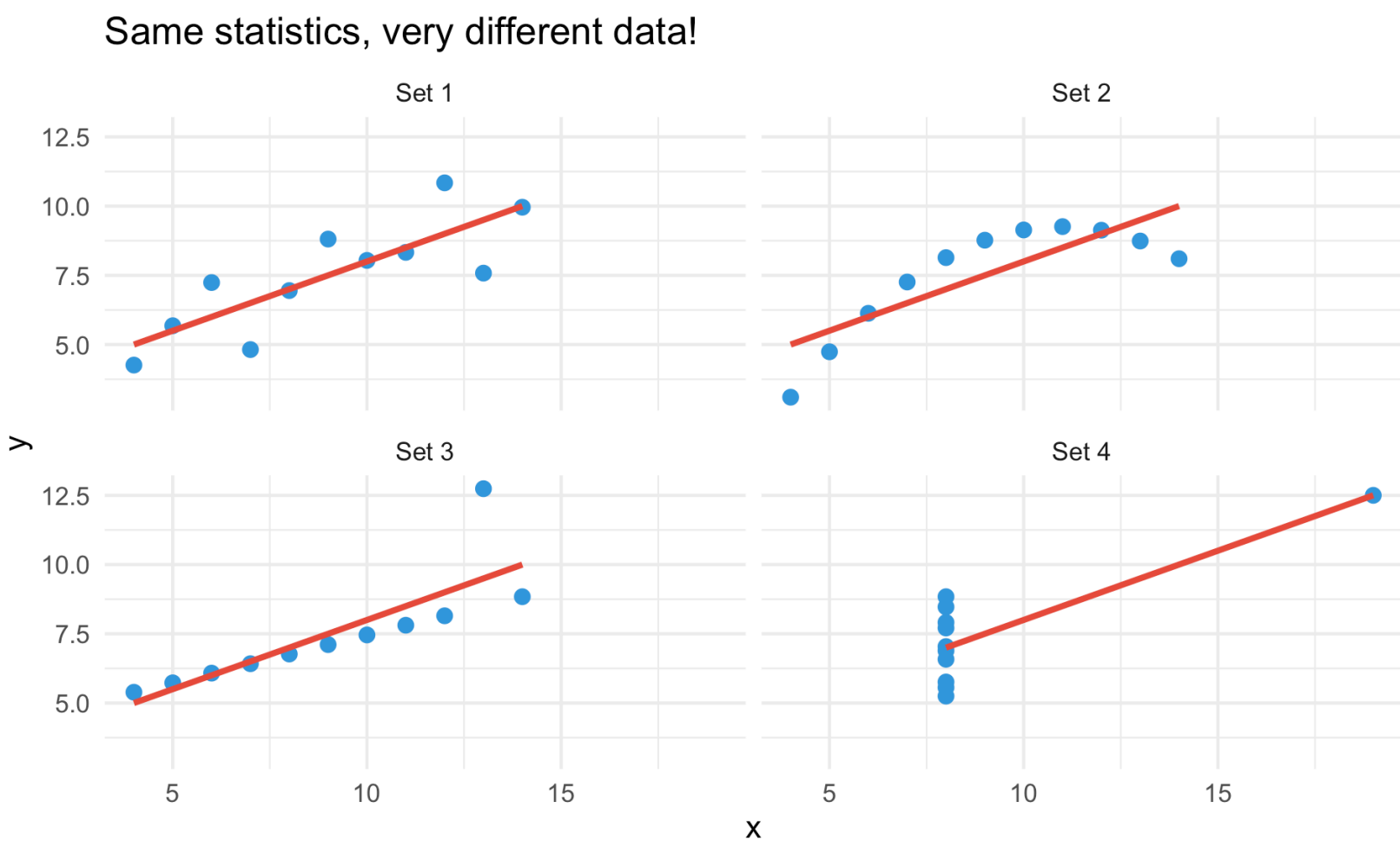
Why visualize?

Anscombe's Quartet

Four datasets with identical summary statistics:

set	mean_x	mean_y	sd_x	sd_y	cor
1	9	7.5	3.32	2.03	0.82
2	9	7.5	3.32	2.03	0.82
3	9	7.5	3.32	2.03	0.82
4	9	7.5	3.32	2.03	0.82

But look at the plots!



Always visualize your data before running statistics.

Introduction to ggplot2

ggplot2 builds plots in layers, like a grammar

- Created by Hadley Wickham (2005)
- Based on the **Grammar of Graphics** by Leland Wilkinson
- Most popular R visualization package
- Part of the tidyverse

The "gg" stands for "Grammar of Graphics"

The grammar of graphics

Every ggplot has three essential components:

1. **Data** — what you want to visualize
2. **Aesthetics (aes)** — how variables map to visual properties
3. **Geoms** — what geometric shapes represent the data

```
# The basic template
ggplot(data = <DATA>, mapping = aes(<MAPPINGS>)) +
  <GEOM_FUNCTION>()
```

Our dataset: mpg

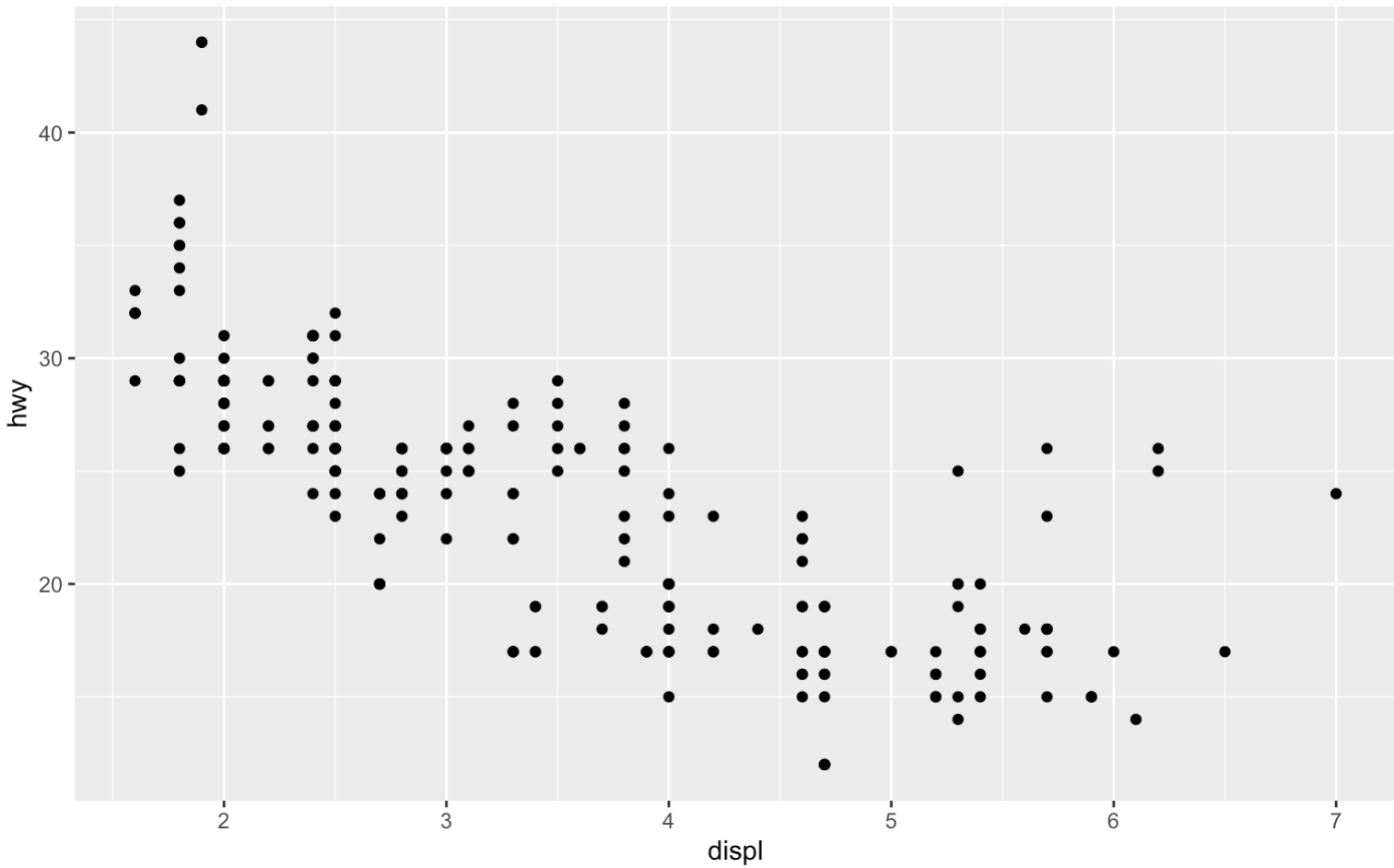
```r
# Fuel economy data for 234 cars
glimpse(mpg)
```

```
Rows: 234
Columns: 11
$ manufacturer <chr> "audi", "audi", "audi", "audi", "audi", "audi", "audi",
"…
$ model        <chr> "a4", "a4", "a4", "a4", "a4", "a4", "a4", "a4 quattro",
"…
$ displ        <dbl> 1.8, 1.8, 2.0, 2.0, 2.8, 2.8, 3.1, 1.8, 1.8, 2.0, 2.0,
2.…
$ year         <int> 1999, 1999, 2008, 2008, 1999, 1999, 2008, 1999, 1999,
200…
$ cyl          <int> 4, 4, 4, 4, 6, 6, 6, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, 8, 8,
…
$ trans        <chr> "auto(l5)", "manual(m5)", "manual(m6)", "auto(av)",
"auto…
$ drv          <chr> "f", "f", "f", "f", "f", "f", "f", "4", "4", "4", "4",
```

Your first plot

```r
# Relationship between engine size and highway mpg
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point()
```

Your first plot



Breaking it down

```r
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  # ↑ Data and aesthetic mappings
  geom_point()
  # ↑ Geometric object (points = scatterplot)
```

- `data = mpg` — use the mpg dataset
- `aes(x = displ, y = hwy)` — map displacement to x, highway mpg to y
- `geom_point()` — represent data as points

You can also write it without the argument names — these are equivalent:

```r
ggplot(mpg, aes(x = displ, y = hwy)) +   # what I'll use in slides
  geom_point()
```

Aesthetic mappings

What are aesthetics?

Aesthetics are visual properties of geoms:

- `x`, `y` — position
- `color` — outline or line color
- `fill` — interior color (bars, boxes)
- `size`, `shape`, `alpha` — size, shape, and transparency

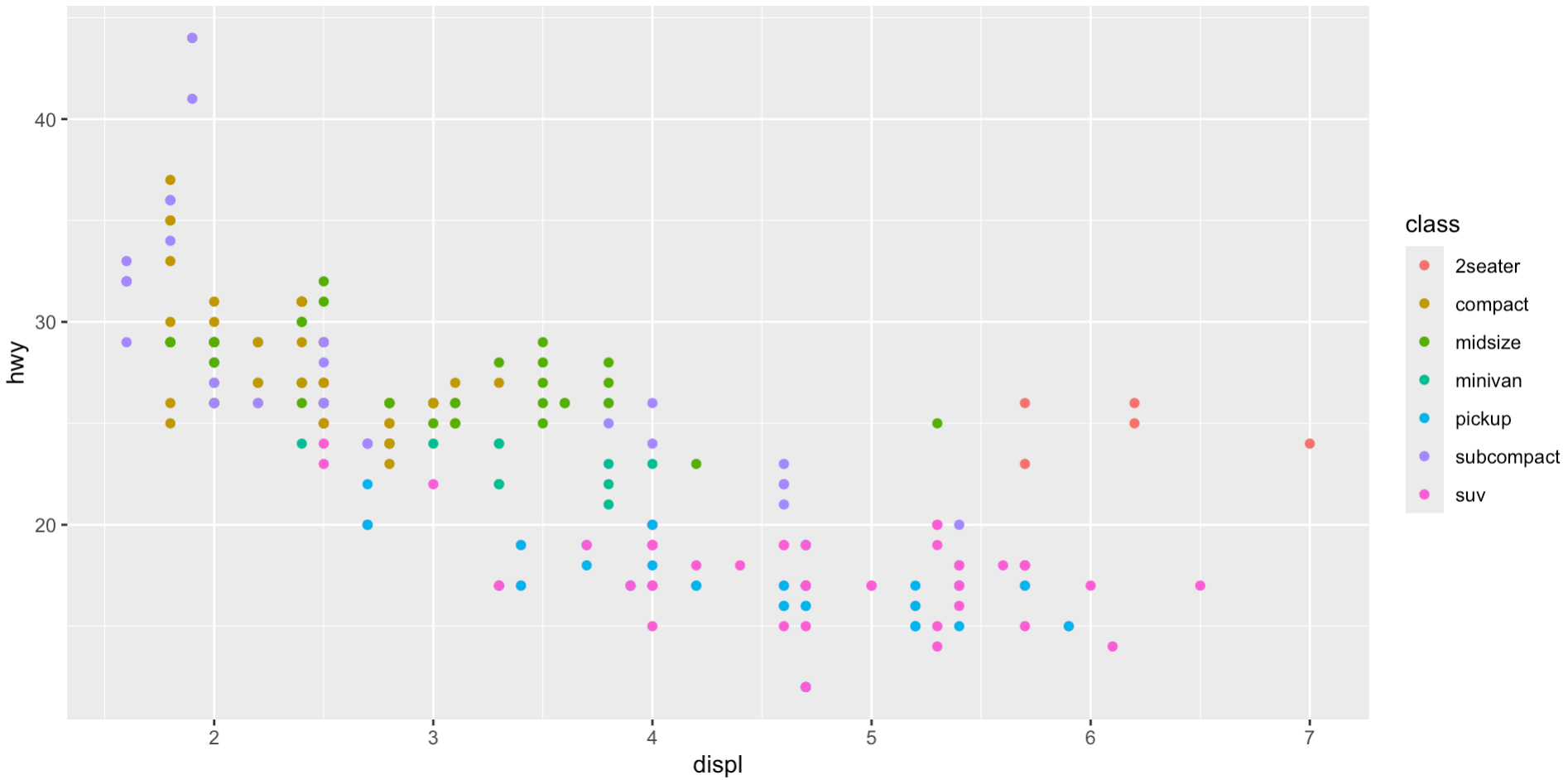
We'll focus on `color` today — you'll explore the others in assignments.

Mapping color to a variable

What if we want to see which points are which car class?

```
ggplot(mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point()
```

Mapping color to a variable



Setting vs. mapping

Mapping — aesthetic varies with data (inside `aes()`)

```
ggplot(mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point()
```

Setting — aesthetic is constant (outside `aes()`)

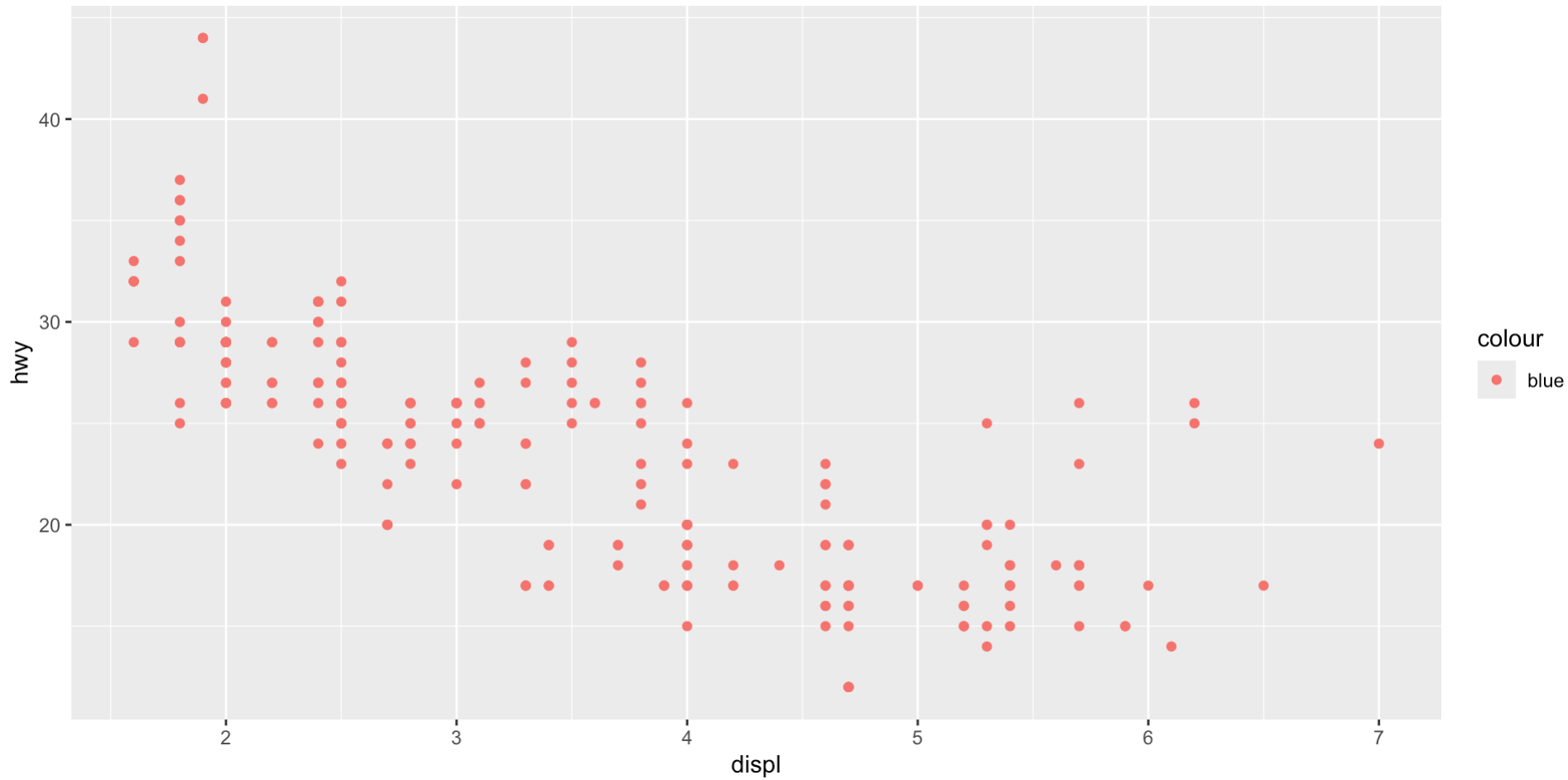
```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(color = "blue", size = 3)
```

Common mistake!

```r
# What happens if you put a constant inside aes()?
ggplot(mpg, aes(x = displ, y = hwy, color = "blue")) +
  geom_point()
```

ggplot thinks “blue” is a category name!

Common mistake!



Different data types need different geoms

Common geoms

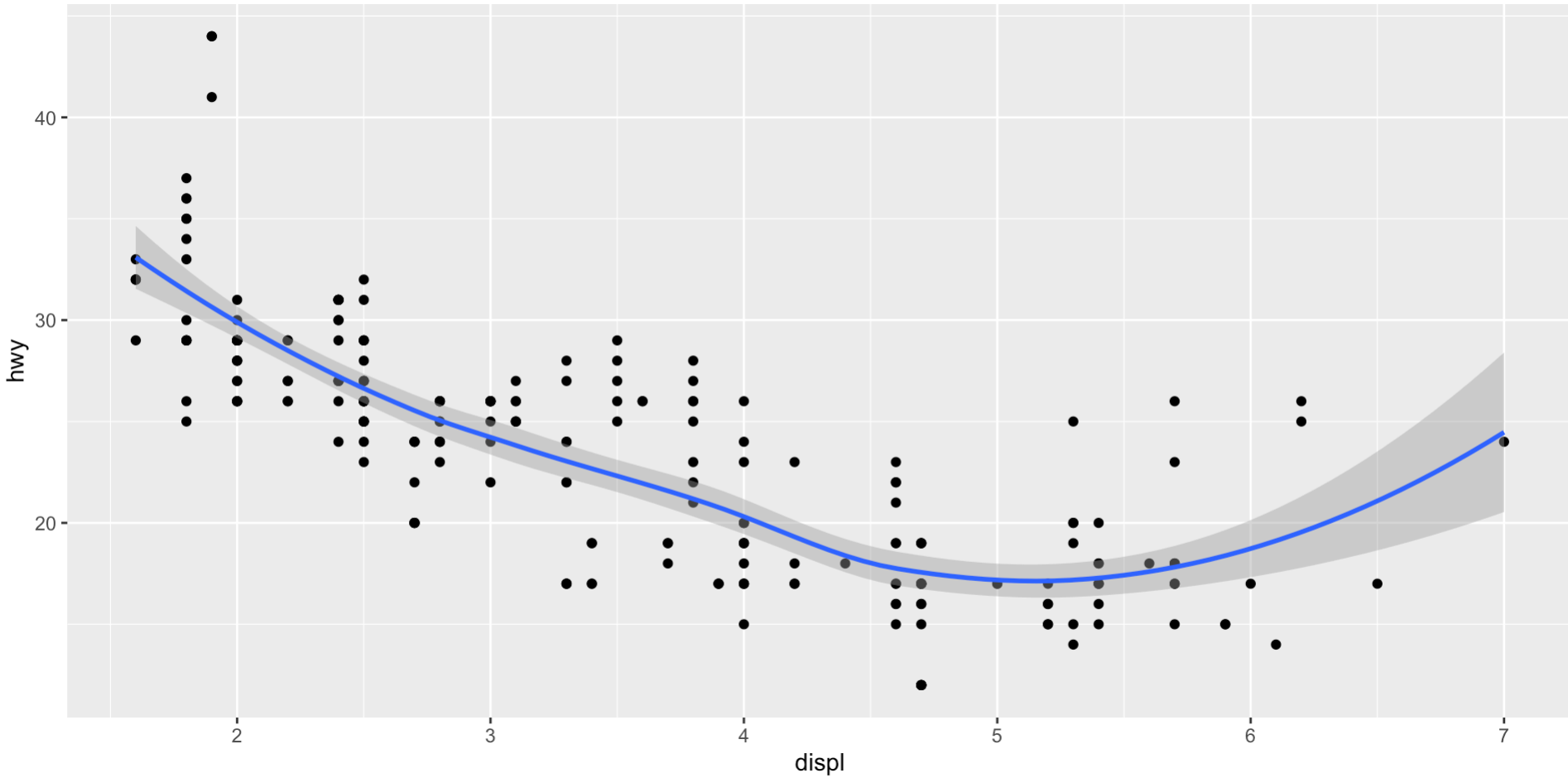
Geom	What it makes
`geom_point()`	Scatterplot
`geom_line()`	Line graph
`geom_bar()`	Bar chart
`geom_histogram()`	Histogram
`geom_boxplot()`	Box plot
`geom_smooth()`	Smoothed line

geom_smooth()

Add a trend line to your scatterplot:

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  geom_smooth()  # Adds a smoothed trend line
```

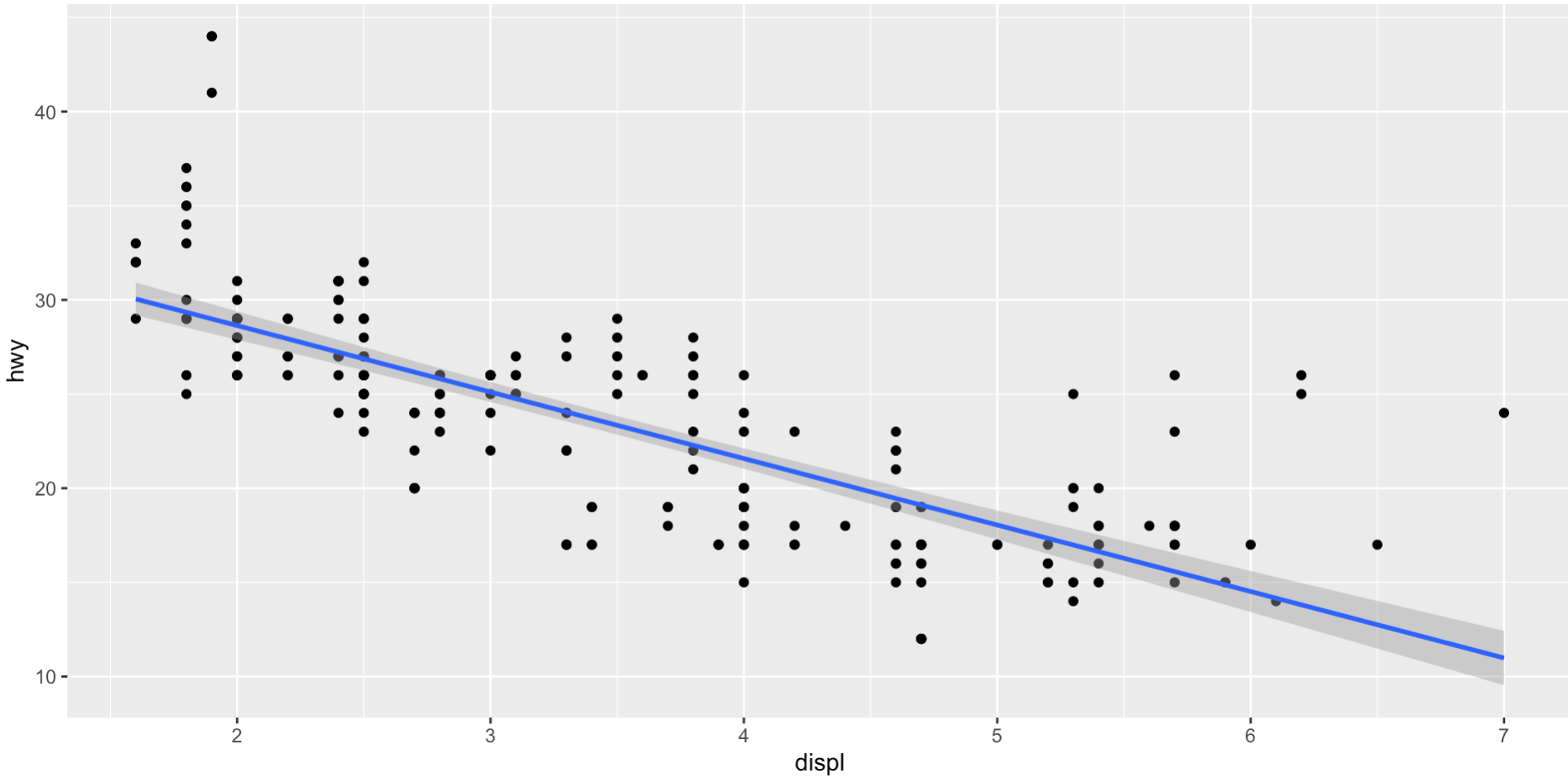
geom_smooth()



Linear trend line

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  geom_smooth(method = "lm")  # lm = linear model
```

Linear trend line

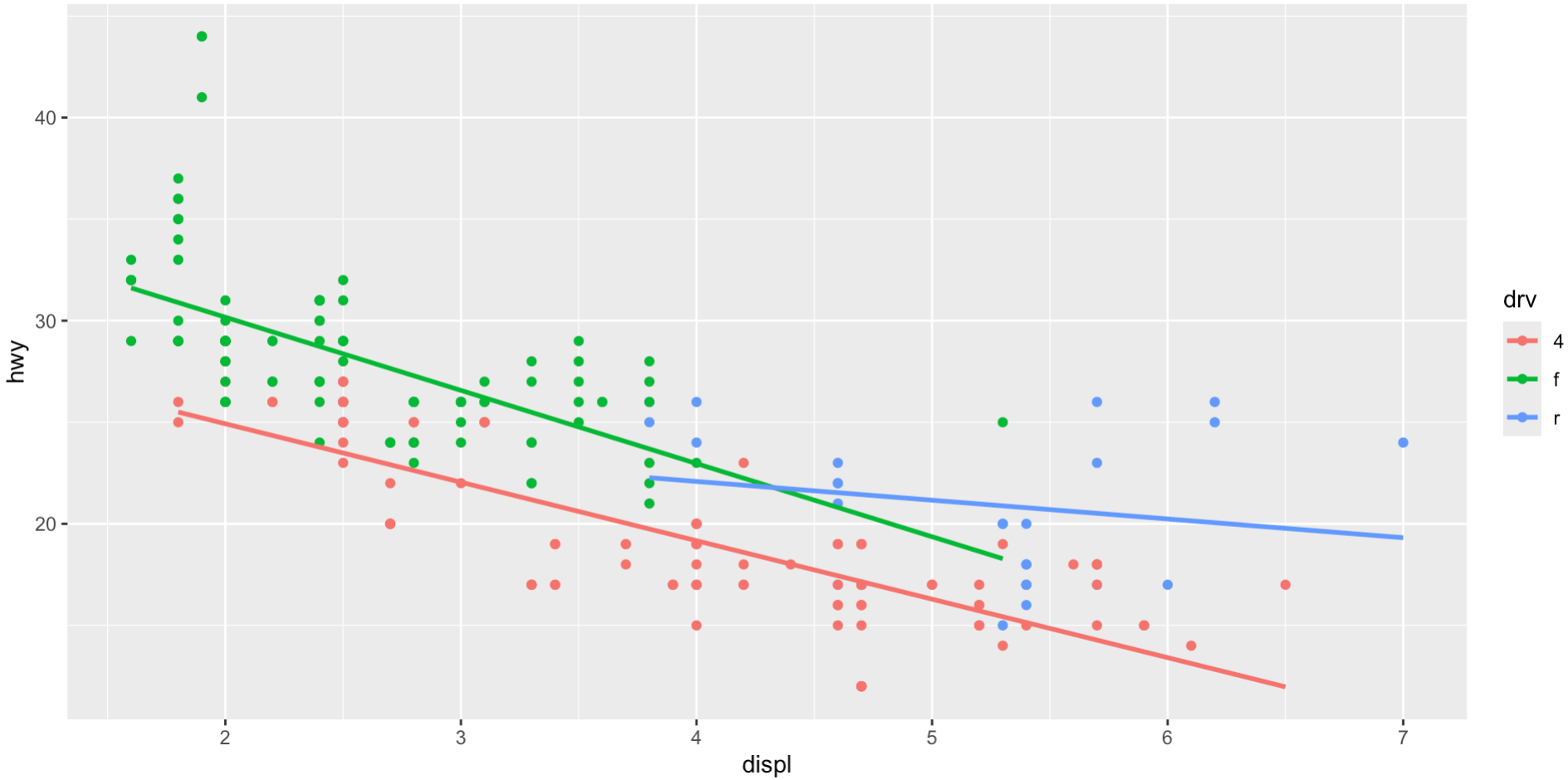


Layering geoms

Each + adds a layer:

```
ggplot(mpg, aes(x = displ, y = hwy, color = drv)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)  # se = FALSE removes confidence ban
```

Layering geoms



Pair coding break

Your turn: 10 minutes

With a partner, create a scatterplot using the `mpg` dataset:

1. Plot `cty` (x-axis) vs `hwy` (y-axis)
2. Color points by **fuel type** (`fl`)
3. Add a smooth trend line
4. Give it a title and axis labels
5. Who can make theirs look the best?

> **Tip**
>
> You have everything you need from the last few slides. Start with the basic template and build from there.

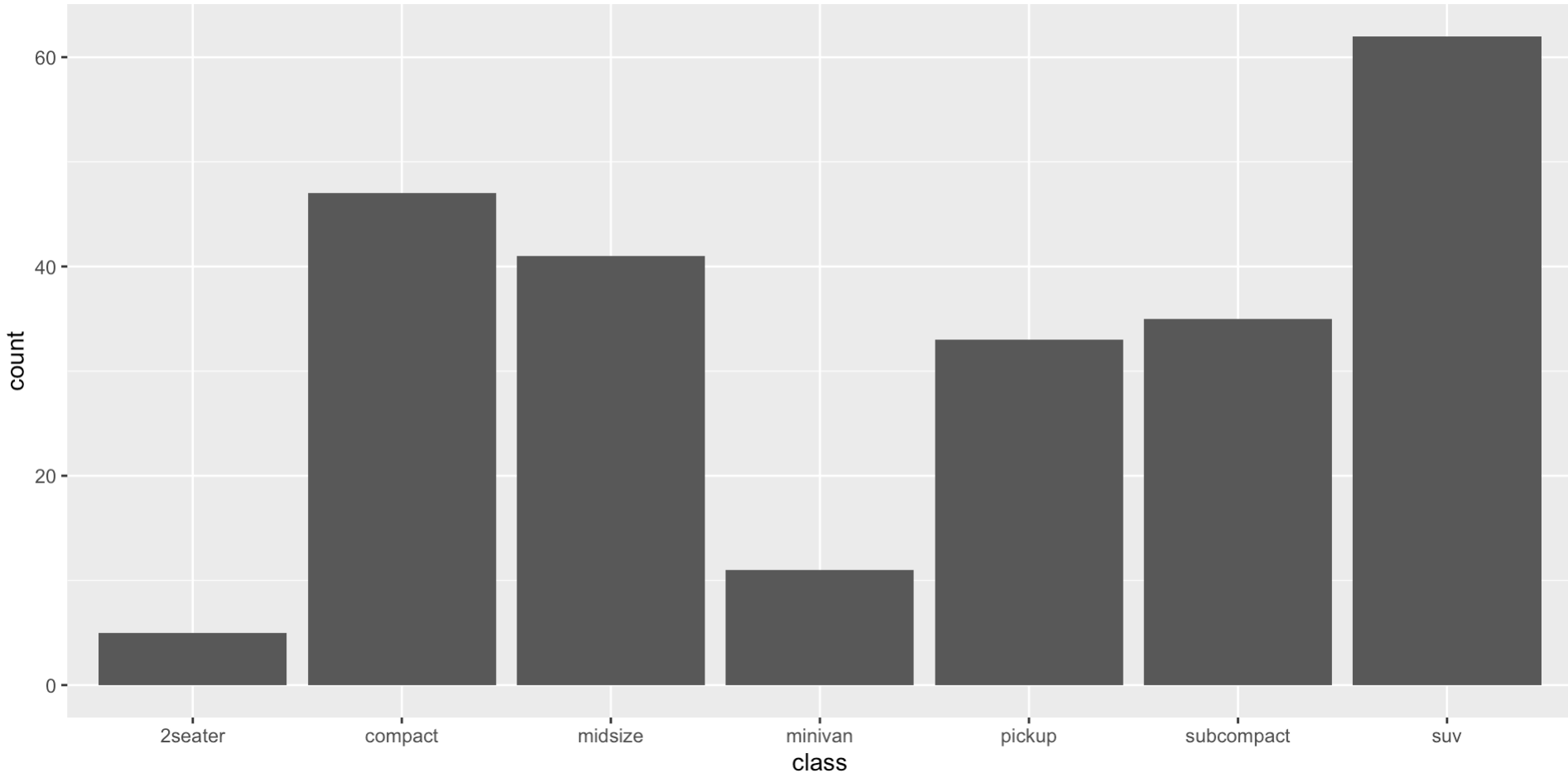
Before we move on

📤 **Upload your code to Canvas** for participation credit. Paste what you have into today's in-class submission — it doesn't need to work perfectly.

geom_bar() — categorical data

```r
# Counts of each car class
ggplot(mpg, aes(x = class)) +
  geom_bar()
```

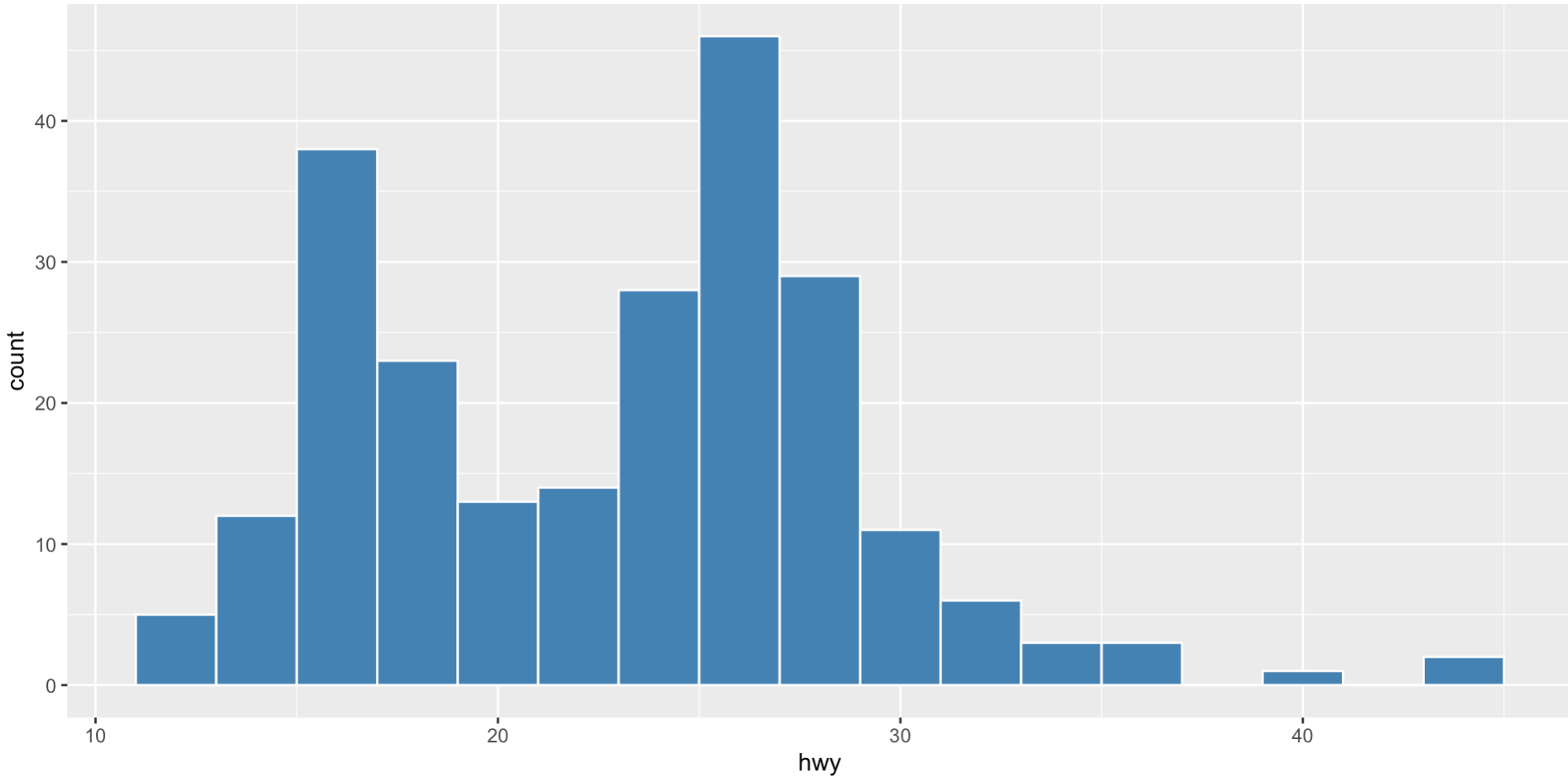
geom_bar() — categorical data



geom_histogram() — distributions

```r
# Distribution of highway mpg
ggplot(mpg, aes(x = hwy)) +
  geom_histogram(binwidth = 2, fill = "steelblue", color = "white")
```

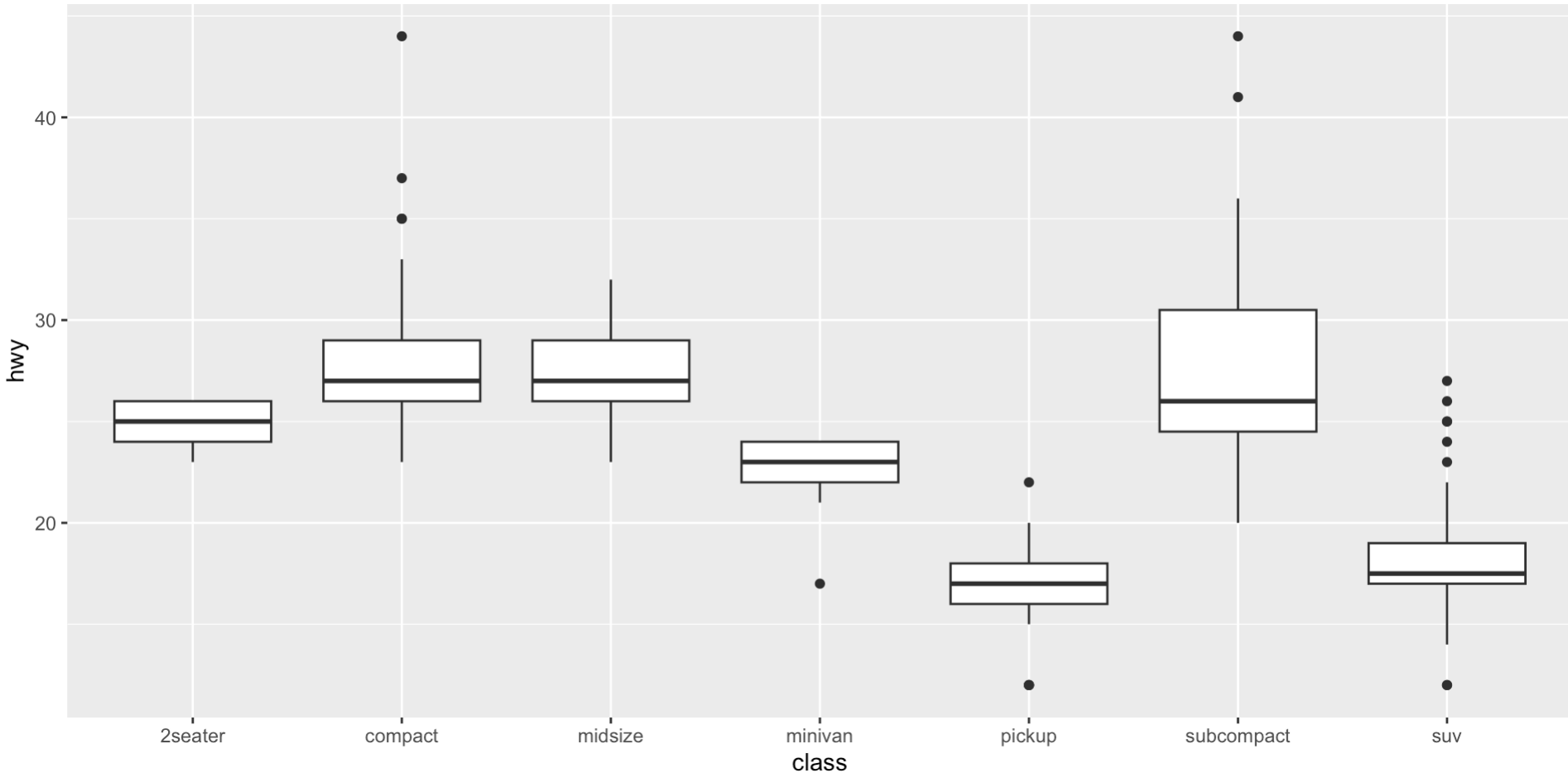
geom_histogram() — distributions



geom_boxplot() — comparing groups

```r
# Highway mpg by car class
ggplot(mpg, aes(x = class, y = hwy)) +
  geom_boxplot()
```

geom_boxplot() — comparing groups



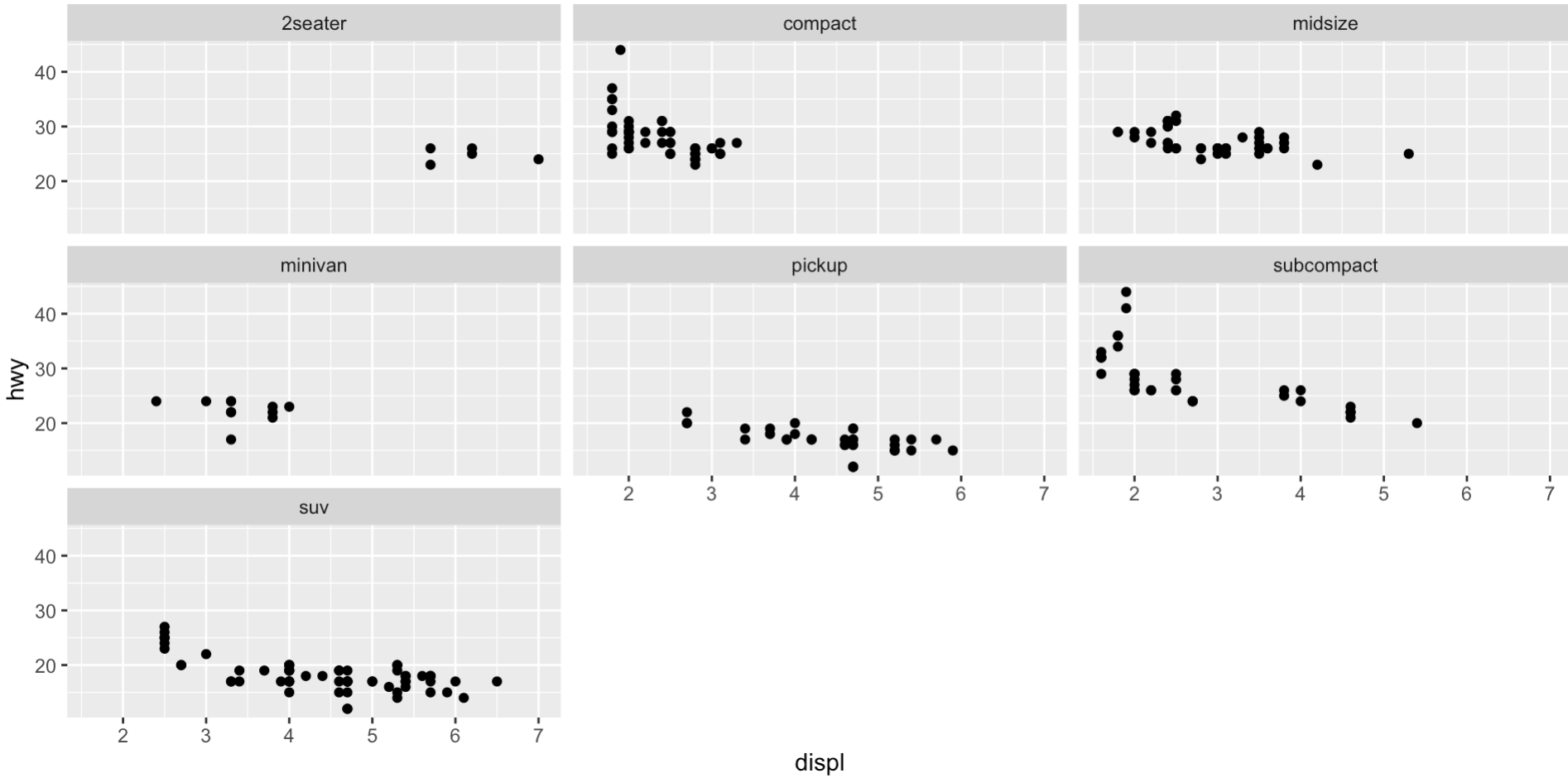
Facets

What are facets?

Facets split your plot into small multiples based on a variable.

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  facet_wrap(~class)
```

What are facets?

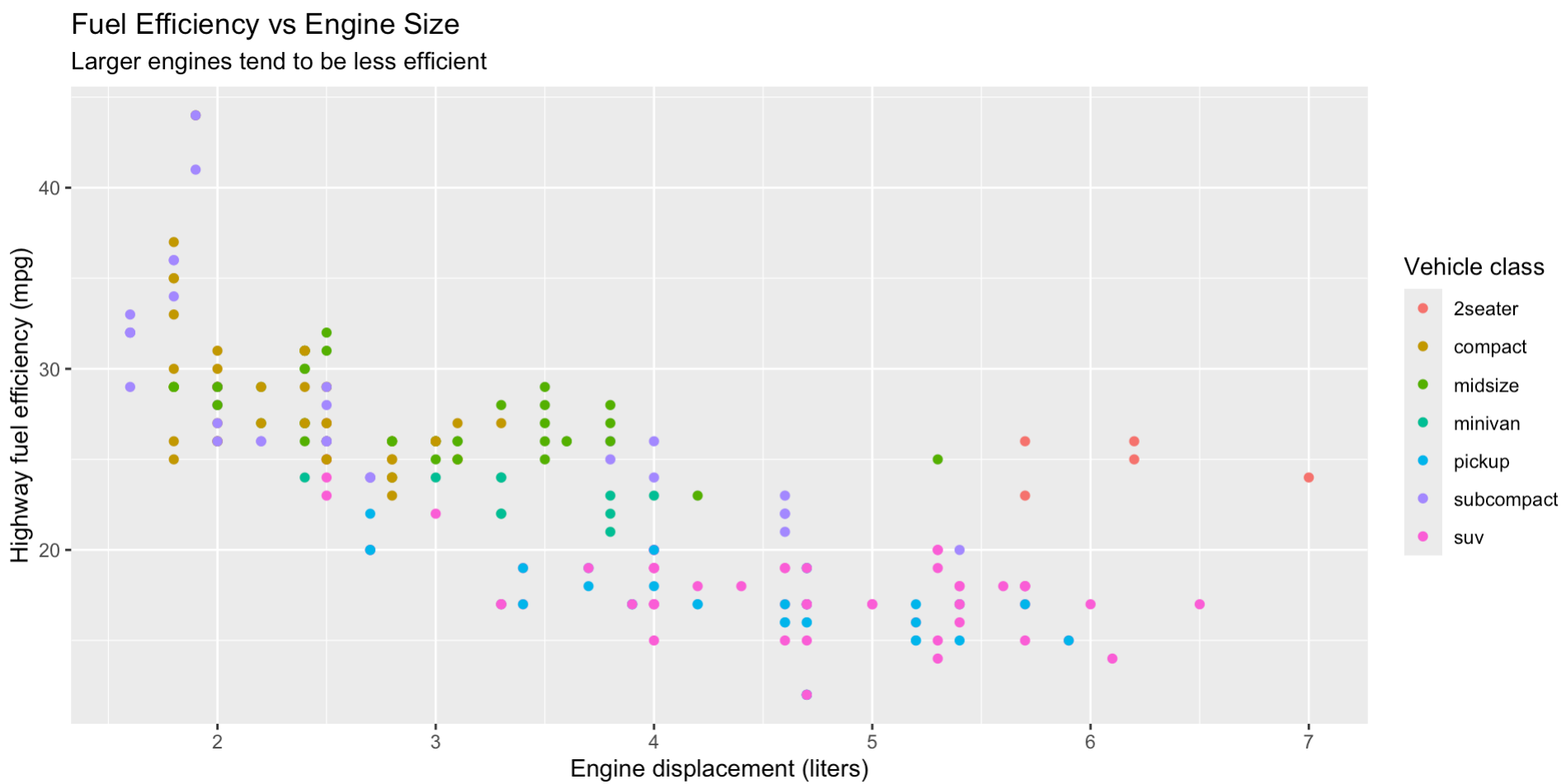


Making it look good

Adding labels

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(aes(color = class)) +
  labs(
    title = "Fuel Efficiency vs Engine Size",
    subtitle = "Larger engines tend to be less efficient",
    x = "Engine displacement (liters)",
    y = "Highway fuel efficiency (mpg)",
    color = "Vehicle class"
  )
```

Adding labels



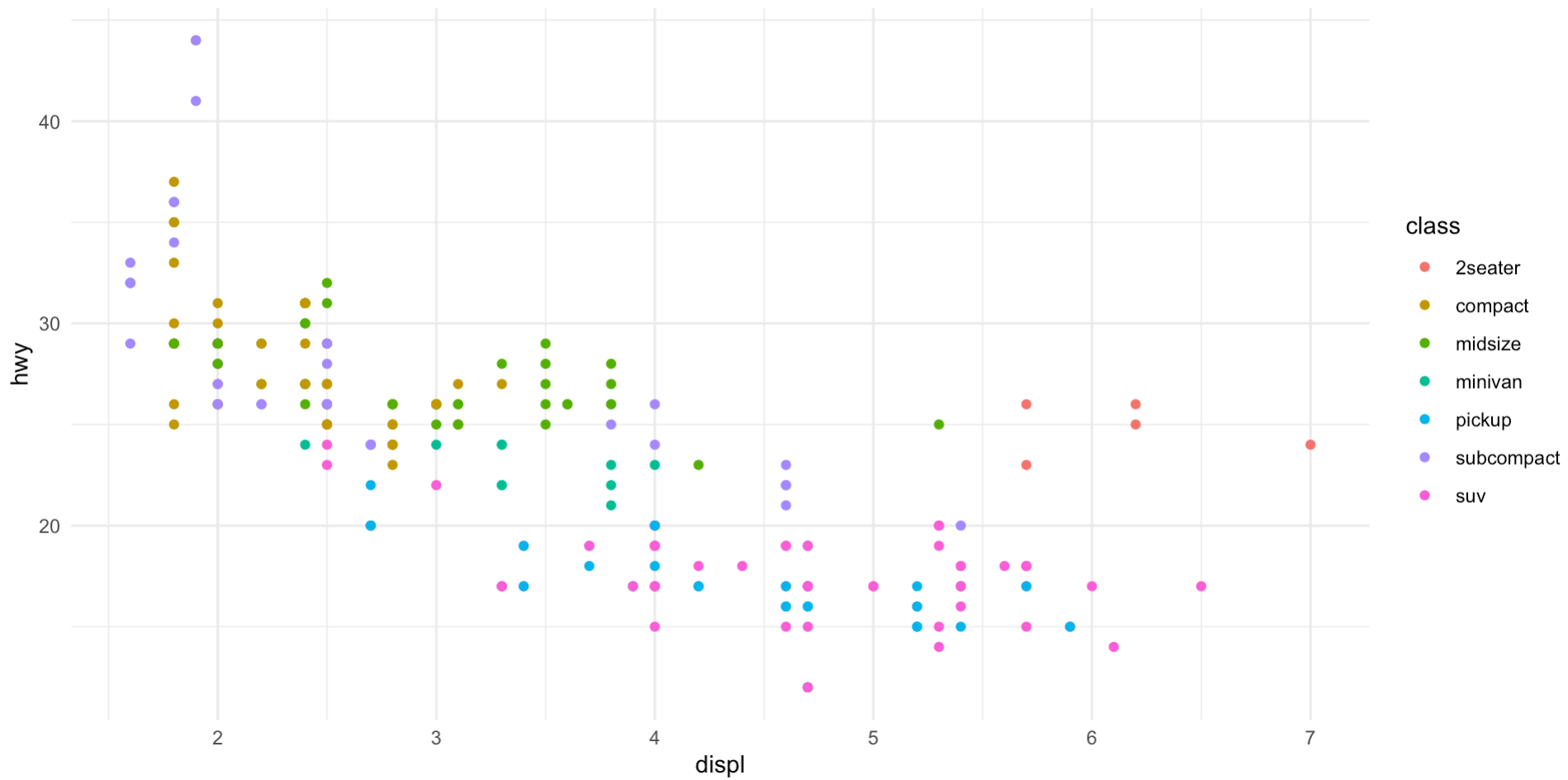
Themes

Themes control the overall look:

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(aes(color = class)) +
  theme_minimal()  # Clean, minimal theme
```

Try different ones: `theme_bw()`, `theme_classic()`, `theme_gray()` (default)

Themes


40
30
20
hwy
2
3
4
5
6
7
displ
class
2seater
compact
midsize
minivan
pickup
subcompact
suv

Saving plots

Use `ggsave()` to save your plot:

```
# Create the plot
my_plot <- ggplot(mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_minimal()

# Save it
ggsave("my_plot.png", my_plot, width = 8, height = 6)
```

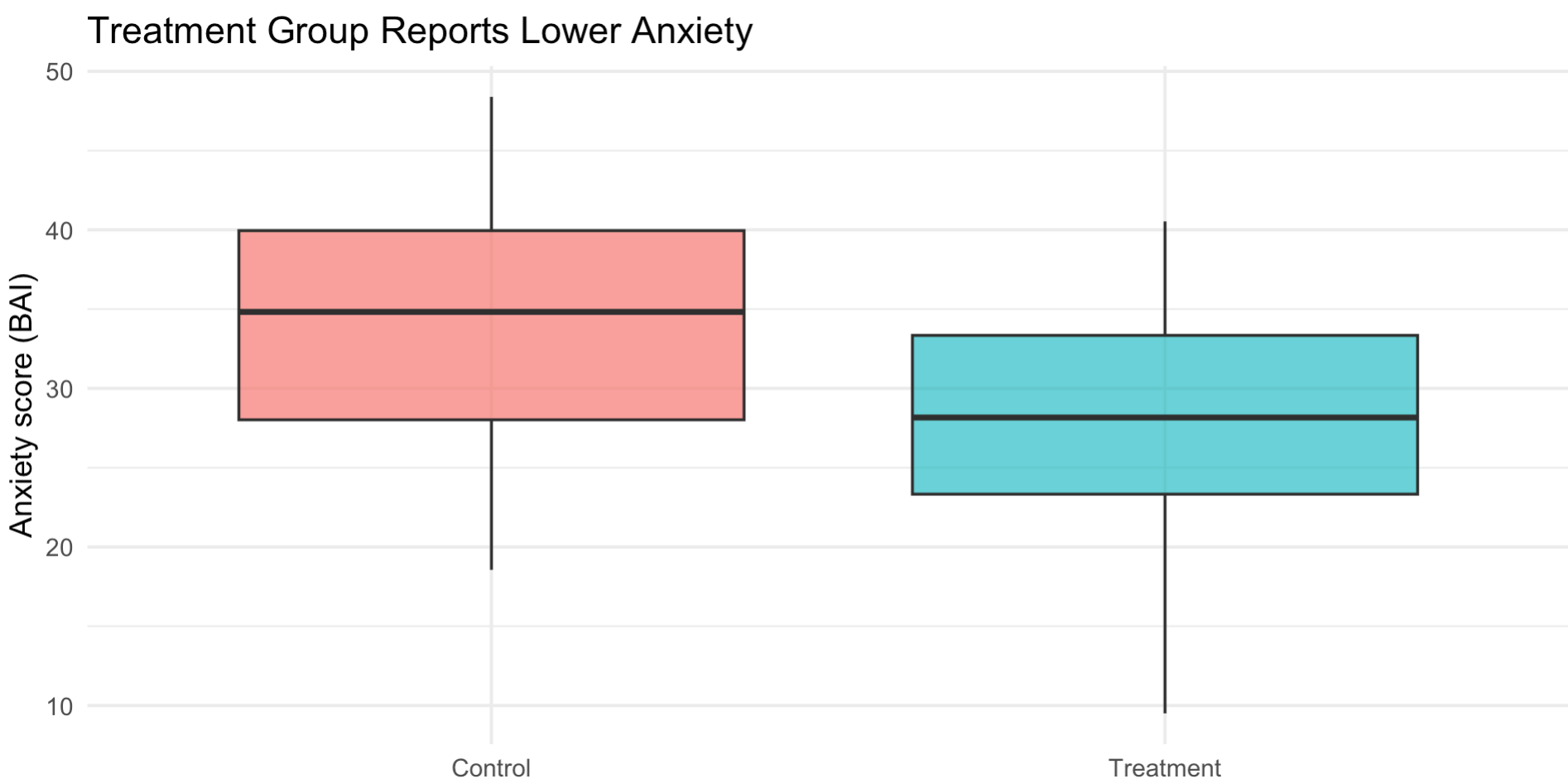
Same template, psychology data

The ggplot template works the same way with any data:

```
set.seed(410)  # Reproducible results
# Simulated experiment: condition vs. anxiety score
psych_demo <- tibble(
  condition = rep(c("Control", "Treatment"), each = 30),
  anxiety = c(rnorm(30, mean = 35, sd = 8), rnorm(30, mean = 28, sd = 8))
)

ggplot(psych_demo, aes(x = condition, y = anxiety, fill = condition)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  labs(
    title = "Treatment Group Reports Lower Anxiety",
    x = NULL,
    y = "Anxiety score (BAI)"
  ) +
  theme_minimal(base_size = 14)
```

Same template, psychology data

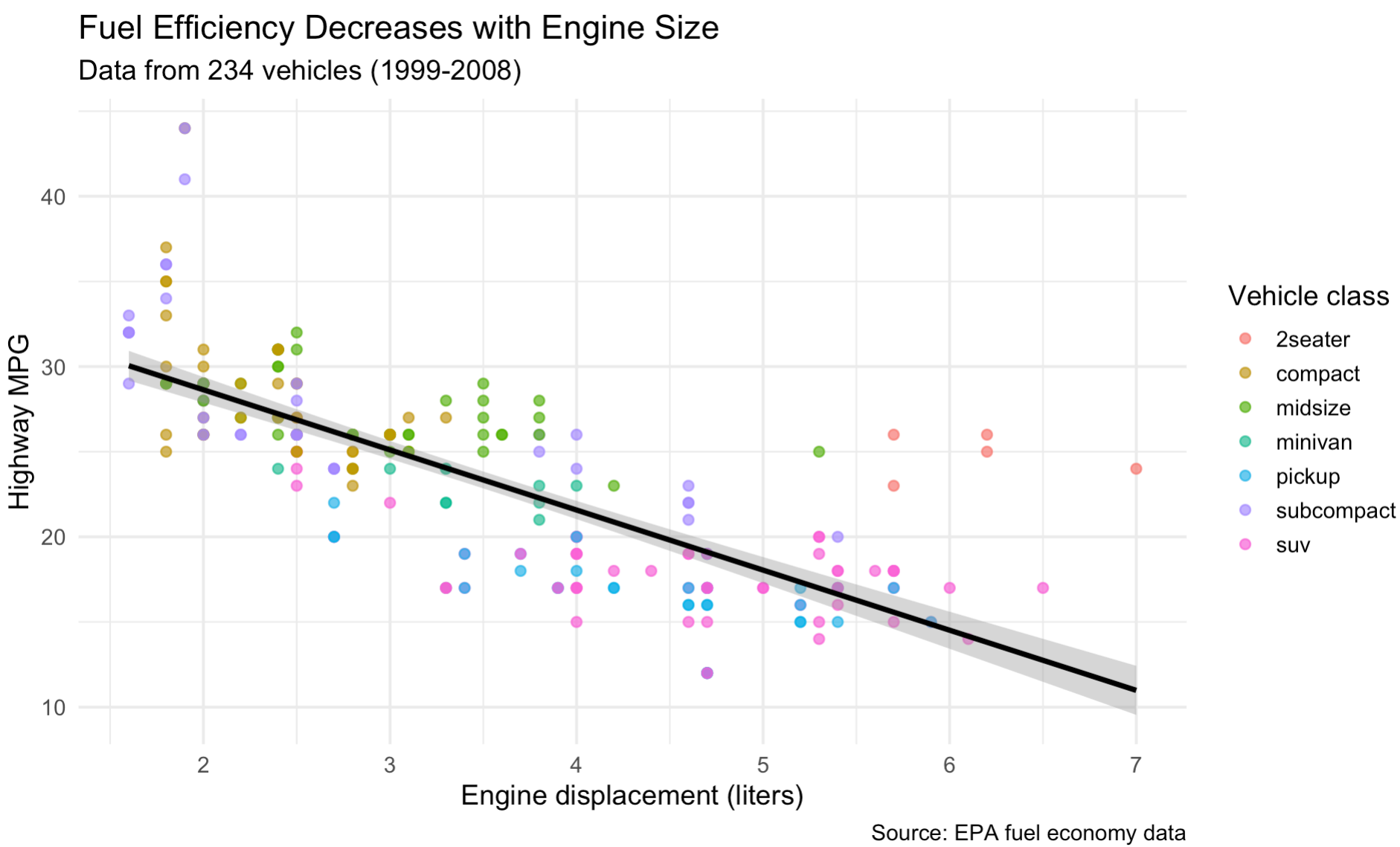


Putting it together

A complete example

```r
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(aes(color = class), size = 2, alpha = 0.7) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(
    title = "Fuel Efficiency Decreases with Engine Size",
    subtitle = "Data from 234 vehicles (1999–2008)",
    x = "Engine displacement (liters)",
    y = "Highway MPG",
    color = "Vehicle class",
    caption = "Source: EPA fuel economy data"
  ) +
  theme_minimal(base_size = 14)
```

A complete example



Get a head start

Assignment 1 preview

Open a new R script in your project. Try these on your own:

1. Make a scatterplot of `displ` vs `hwy` from `mpg`
2. Color it by `class`
3. Facet by `drv` (drive type)
4. Save it with `ggsave()`

This is the first part of Assignment 1.

Solution

Wrapping up

Before next class

📖 **Read:** R4DS Ch 3: Data transformation (sections 3.1–3.4)

📝 **Assignment 1** is due Sunday at 11:59 PM

Never trust a summary statistic you haven't plotted

Today's template — you'll use it all quarter:

```
ggplot(<DATA>, aes(<MAPPINGS>)) +
  <GEOM_FUNCTION>() +
  labs(<LABELS>) +
  theme_<THEME>()
```

Next time: Data Transformation with dplyr